



## PROJET

**Title:** RBP-Score: An integrative bioinformatic pipeline for the identification of phage Receptor Binding Proteins

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### Introduction:

Bacteriophages (phages) are the most abundant biological entities on Earth and display a remarkable specificity towards their bacterial hosts, often at the species or even strain level. This specificity is largely determined by Receptor Binding Proteins (RBPs), structural components of the phage particle that mediate recognition and binding to specific receptors on the bacterial cell surface. Due to their central role in host recognition, RBPs hold enormous potential for biotechnological applications, including phage engineering, bacterial diagnostics and the development of alternative strategies to combat antibiotic-resistant pathogens.

Despite the rapid increase in the number of available phage genome sequences, the identification and annotation of RBPs remains a major bioinformatic challenge. RBPs are highly diverse and typically show low sequence conservation, which severely limits the performance of traditional homology-based approaches. However, recent studies indicate that RBPs may retain conserved structural features and functional modularity, even in the absence of detectable sequence similarity. The availability of accurate protein structure predictions through AlphaFold has opened new possibilities for structure-based protein identification.

This project aims to address this challenge by developing a curated reference dataset of experimentally validated phage RBPs and by implementing an integrative, reproducible bioinformatic pipeline to identify and prioritise RBPs in phage genomes using sequence-, phylogeny- and structure-based evidence.

### Aims:

To manually curate a high-quality database of phage RBPs with clear experimental validation reported in the literature.

To identify conserved sequence motifs, domains and/or structural features shared among validated RBPs and assess their discriminatory power.

To develop a fully reproducible bioinformatic pipeline (e.g. in Snakemake) for the analysis of candidate RBPs.

To assign independent scores to each candidate protein based on: sequence homology, phylogenetic relatedness, structural similarity using AlphaFold-predicted models.

To integrate the individual scores into a weighted global score reflecting the probability of a protein being a bona fide RBP.

To benchmark and critically evaluate the performance of the individual approaches and the integrated scoring strategy using the curated RBP dataset as a reference.